SOFTWARE ENGINEERING

Lesson 2 People Metrics

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INTRODUCTION TO PEOPLE MANAGEMENT

- Hiring and maintaining a senior team to solve **insufficient Team Skills and allow you to grow**
- Offer competitive conditions of salary, and a **great employee experience** focused on the workplace culture of wellness, security and **career for engineers**
- Keeping teams o**n the same page beside** the ton of information that they receive
- Improve defining the **goals and objectives and** reducing miscommunication **Cause Conflicts**
- Deal with **unrealistic deadlines** and **balance the expectations** of the organization
- Finding the right **Project Management Software Model** to minimize risks and control **scope creep**
- Plan, adapt, solve problems, and manage resources to guide the SaaS company **while finding product-market fit**
- Setting the ground for better, **more reliable, and faster collaboration**

IMPORTANCE OF HIGH-PERFORMANCE ENGINEERING IN SCALING A BUSINESS

- A short list of how a senior engineering team will help your company to save money:
- Senior engineers will identify product-market fit and product performance issues in the early stages saving time and resources
- Your company gain more flexibility and time to optimize software performance and improve processes to scale your Software Development Method
- You finally will be able to automate many development processes that don’t make sense to be done manually anymore
- The quality of the product will help your company to acquire customers more consistently
- More efficiency means less maintenance or eliminates the need for any additional technology investments to solve problems
- Senior engineers create software for more sustainable and long-term products increasing the return on investments (ROI)

KEEPING THE SOFTWARE ENGINEERS AT THEIR PRIME

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IMPORTANCE OF HIGH-ENGINEERING IN SCALING A BUSINESS

1. HIRING. Get specialist's support to plan and attract senior engineers

- Get a specialist in tech recruitment to support you in Develop Recruiting Strategies, Identify Recruiting Sources and Networks, and improve the interview process
- Use the IT Staff Augmentation and nearshore strategy for hiring and scaling the Software Development Team to cover skill gaps after evaluating the existing staff and identifying a lack of experience to fill.
- Doesn't just sit there, but go hunt for your new Software Engineers, wherever they are
- Asses proficiency, skills requirements, as well as the product roadmap
- Analyze non-tech skills in addition to technical ones (communication, decision-making, attention to detail, presentation, negotiation skills, etc).
- Set up a strategy to identify behavioral, social, and collaboration traits
- Take time enough to final approvals
- Organize and execute the offer/hire documentation: mainly the Statement of Work (SOW) and Master Services Agreement (MSA)

INTRODUCTION TO PEOPLE METRICS IN SOFTWARE ENGINEERING

- The concept of people metrics originated in the area of HR management.
- It provides a quantifiable measure of people's activity.
- These metrics are helpful to provide evidence of performance against things like objectives and goals.
- It has become important for Balanced Scorecards and other performance measurement systems.
- The reason is thanks to the necessity for effective management over human resource capital; i.e. the proper utilization of the intellectual capital that helps in the derivation of value.
- To make matters more urgent, senior management sometimes fails to utilize the actual value of the human resources within an organization
- It becomes very important to explain or we can say demonstrate the value of people capital to executable management.
- People's metrics can provide evidence of performance against objectives and goals.

INTRODUCTION

People metrics in software engineering refer to the measurement and tracking of various aspects of a software development team's performance and behavior. These metrics can include data on team member productivity, code quality, team morale, and employee turnover rate. The goal of using people metrics is to gain insights into the team's performance and identify areas for improvement. This can help managers make data-driven decisions to improve the overall performance of the team and the software development process.

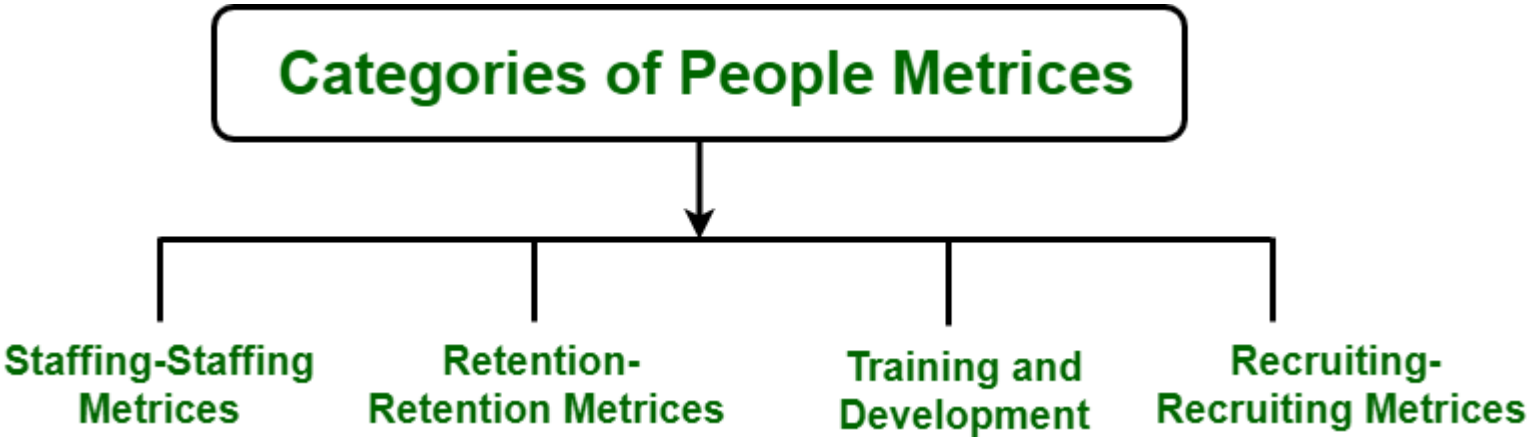
Following are some examples of people metrics which may include:

- Calculation of average to fill a new position.
- New hire failure factor.
- Headcount by gender.

IMPORTANCE

This also assists HR in speaking the language of the chief team. By measuring people management activities, the worth of business outcomes is often translated into quantifiable evidence and presented in a language that resonates with business leaders.

CATEGORIES OF PEOPLE METRICES



Staffing Metrics: Staffing metrics are a type of people metric that focuses on the size and composition of a software development team. This includes information and data.

This type of metric quantifies the return on investment in your employees. These measures include quantities such as :

- Cost per Time
- Recruiting Efficiency Ratio
- Cost to Replace an Employee

Retention Metrics: Retention metrics are a type of people metric that focuses on the rate at which employees are leaving a software development team.

It often measures important aspects of turnover. Management often wishes to quantify such variables:

- Turnover Rate
- Average Tenure
- The Rate of Veteran Workers
- The Financial Impact of Employee Turnover

Results often indicate what proportion each separating employee is costing the corporation and help the corporation create proactive plans to stop the loss of top talent.

Training and Development Metrics: Training and development metrics are a type of people metric that focuses on the education and skill development of a software development team.

It quantifies the learning processes of new employees. These measures include activities such as:

- Orientation
- Training Process Time and Costs
- The time and cost of on-the-job learning

Results often demonstrate the success of professional development processes and how much they help the organization achieve its business goals.

Recruiting Metrics: Recruiting metrics are a type of people metric that focuses on the effectiveness of the recruiting process for a software development team.

It measures activities involved in the stages of attracting and selecting top talent. Decision-makers frequently want to quantify variables such as :

- New Hire Performance
- Turnover Rates of New Hires
- Impact of a Poor Hire
- Return on Investment in a New Hire

People metrics in software engineering refer to the quantitative and qualitative measures of the performance, productivity, and satisfaction of the people involved in the software development process, including developers, testers, project managers, and other stakeholders. People metrics are used to assess the efficiency and effectiveness of the software development process and identify areas for improvement.

Objective metrics and Subjective metrics.

Objective metrics are quantitative measures of the performance and productivity of individuals and teams in the software development process. Examples of objective metrics include:

1. Lines of code (LOC): A measure of the size of the software system in terms of the number of lines of code written.
2. Function points (FP): A measure of the functionality of the software system in terms of the number of inputs, outputs, inquiries, and files processed.
3. Defect density: A measure of the quality of the software system in terms of the number of defects per unit of code.
4. Cycle time: A measure of the time it takes to complete a specific task or activity in the software development process, such as coding, testing, or deployment.

5. Effort: A measure of the resources, such as time, money, and people, required to complete a specific task or activity in the software development process.
6. Subjective metrics are qualitative measures of the satisfaction and engagement of individuals and teams in the software development process. Examples of subjective metrics include:
7. Job satisfaction: A measure of the level of satisfaction and motivation of individuals and teams in their work in the software development process.
8. Team cohesion: A measure of the level of collaboration, communication, and trust among team members in the software development process.
9. Communication effectiveness: A measure of the effectiveness and clarity of communication among team members in the software development process.

10. Learning and development: A measure of the opportunities for learning, growth, and development of individuals and teams in the software development process.